# Same Position, Two Look-Up Tables:

Novice (square → square) vs. Master (pattern → chunk)  [Chase & Simon, 1973]

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.

**Board flashed for 5 seconds**

~25 pieces from a real game,

then reconstruct from memory

piece by piece

pattern chunks

## Novice Player

**Novice Look-Up Table**
one row per single piece

| Isolated Piece | Memory Slot |
|---|---|
| White pawn on e4 | Slot 1 |
| Black knight on f6 | Slot 2 |
| White bishop on c4 | Slot 3 |
| Black pawn on d6 | Slot 4 |
| ... ~25 separate pieces | ... slots run out (~7) |

**Outcome**

Only ~6 pieces recalled;

working memory overflows

## Chess Master

**Master Look-Up Table**
one row per recognized chunk

| Recognized Pattern | Stored as One Chunk |
|---|---|
| Castled king + pawn shield (g1,f2,g2,h2) | "Kingside castle" = 1 chunk |
| Ruy Lopez opening structure | "Ruy Lopez" = 1 chunk |
| Isolated queen-pawn formation | "IQP" = 1 chunk |
| ... a handful of familiar gestalts | ... a few big chunks |

**Outcome**

Nearly the whole board recalled

(edge vanishes on random boards)